

DBMS - PAYMENT MANAGEMENT SYSTEM

Title - Flipkart E-Commerce Project

1. Database Design

1.1 Payment Table

Entity:Payment

Attribute	Key type
Payment_ID	Primary key
Order_ID	Foreign key
Payment_Method	-
Payment_Status	-
Payment_Date	-
Amount	-

2. Table Relationships

Orders – Payment

- Relationship: One-to-Many (1:M)
- One order can have multiple payment transactions.
- Each payment is associated with one order.
- Order_ID is the foreign key in the Payment table.

3. Table Creation

3.1 Create Payment Table

```
SQL> CREATE TABLE Payment (  
2     Payment_ID NUMBER PRIMARY KEY,  
3     Order_ID NUMBER,  
4     Payment_Mode VARCHAR2(20) NOT NULL,  
5     Payment_Date DATE NOT NULL,  
6     Payment_Amount NUMBER(10,2) NOT NULL,  
7     Payment_Status VARCHAR2(20) NOT NULL,  
8     FOREIGN KEY (Order_ID) REFERENCES Orders(Order_ID)  
9 );
```

Table created.

4. Payment Data Insertion

```
SQL> INSERT INTO Payment
  2  VALUES (601, 401, 'UPI', SYSDATE, 55000, 'Success');

1 row created.

SQL>
SQL> INSERT INTO Payment
  2  VALUES (602, 402, 'Card', SYSDATE, 25000, 'Success');

1 row created.

SQL>
SQL> INSERT INTO Payment
  2  VALUES (603, 403, 'UPI', SYSDATE, 1598, 'Failed');

1 row created.

SQL>
SQL> INSERT INTO Payment
  2  VALUES (604, 404, 'Cash', SYSDATE, 2500, 'Success');

1 row created.

SQL>
SQL> INSERT INTO Payment
  2  VALUES (605, 405, 'Net Banking', SYSDATE, 650, 'Failed');

1 row created.
```

5. Manage Payment Transactions

5.1 Display Successful Payments

```
SQL> SELECT *
  2  FROM Payment
  3  WHERE Payment_Status = 'Success';
```

PAYMENT_ID	ORDER_ID	PAYMENT_MODE	PAYMENT_D	PAYMENT_AMOUNT
601	401	UPI	29-SEP-26	55000
602	402	Card	29-SEP-26	25000
604	404	Cash	29-SEP-26	2500

5.2 Display Failed Payments

```
SQL> SELECT *
  2  FROM Payment
  3  WHERE Payment_Status = 'Failed';
```

PAYMENT_ID	ORDER_ID	PAYMENT_MODE	PAYMENT_D	PAYMENT_AMOUNT
603	403	UPI	29-SEP-26	1598
605	405	Net Banking	29-SEP-26	650

6. Modify Payment Status

```

SQL> UPDATE Payment
  2 SET Payment_Status = 'Success'
  3 WHERE Payment_ID = 603;

1 row updated.

SQL>
SQL> COMMIT;

Commit complete.

SQL> SELECT *
  2 FROM Payment
  3 WHERE Payment_ID = 603;

```

PAYMENT_ID	ORDER_ID	PAYMENT_MODE	PAYMENT_D	PAYMENT_AMOUNT
603	403	UPI	29-SEP-26	1598
Success				

7. Analyze Payment Methods

```

SQL> SELECT
  2     Payment_Mode,
  3     COUNT(*) AS Total_Transactions
  4 FROM Payment
  5 GROUP BY Payment_Mode
  6 ORDER BY Payment_Mode;

```

PAYMENT_MODE	TOTAL_TRANSACTIONS
Card	1
Cash	1
Net Banking	1
UPI	2

Calculate amount collected by payment mode

```
SQL> SELECT
  2     Payment_Mode,
  3     SUM(Payment_Amount) AS Total_Amount
  4 FROM Payment
  5 WHERE Payment_Status = 'Success'
  6 GROUP BY Payment_Mode
  7 ORDER BY Payment_Mode;
```

PAYMENT_MODE	TOTAL_AMOUNT
Card	25000
Cash	2500
UPI	56598

8. Payment Transaction Report

```
SQL> SELECT
  2     p.Payment_ID,
  3     c.First_Name || ' ' || c.Last_Name AS Customer_Name,
  4     p.Order_ID,
  5     p.Payment_Mode,
  6     p.Payment_Date,
  7     p.Payment_Amount,
  8     p.Payment_Status
  9 FROM Payment p
 10 JOIN Orders o
 11     ON p.Order_ID = o.Order_ID
 12 JOIN Customer c
 13     ON o.Customer_ID = c.Customer_ID
 14 ORDER BY p.Payment_ID;
```

PAYMENT_ID	CUSTOMER_NAME	ORDER_ID	PAYMENT_MODE	PAYMENT_D	PAYMENT_AMOUNT	PAYMENT_STATUS
601	Renu Srinivasan	401	UPI	29-SEP-26	55000	Success
602	Arjun Kumar	402	Card	29-SEP-26	25000	Success

603
Priya Kumar
403 UPI 29-SEP-26 1598 Success

604
Rahul Sharma

PAYMENT_ID

CUSTOMER_NAME

ORDER_ID	PAYMENT_MODE	PAYMENT_D	PAYMENT_AMOUNT	PAYMENT_STATUS
404	Cash	29-SEP-26	2500	Success

605
Keerthana Mohan
405 Net Banking 29-SEP-26 650 Failed